Introduction to mathematics for economists

Exercise Sheet #1

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During the tutorials, we will solve exercises 1, 3, 5, 8, 10, and 12. So you must prioritize these. You will be provided a full answer sheet. It's normal to struggle with (at least) some exercises, don't panic, ask question and solve exercises with your classmates.

Continuity

Exercise 1 : ε-δ proofs

1. Show that $f(x) = 3x + 2$ is continuous at every $x_0 \in \mathbb{R}$ using the ε-δ definition. *(Hint: just ask yourself what δ you must pick)*
2. Show that $g(x) = \frac{1}{x}$ is continuous at every $x_0 > 0$. *Note: this question is a bit hard, don't hesitate to collaborate.*
3. Show that $g(x) = \frac{1}{x}$ is **not** uniformly continuous on $(0, \infty)$. *(Hint: you may depart from the $\epsilon - \delta$ style and find two sequences u_n and v_n such that $|u_n - v_n| \to 0$ but $|f(u_n) - f(v_n)|$ does not converge to 0.)*
4. Show that g **is** uniformly continuous on $[a, \infty)$ for any fixed $a > 0$. Explain why being bounded away from 0 matters in this case.

Exercise 2 : Sequential continuity and a demand function

In economics, a **demand function** $D(p)$ maps a price $p > 0$ to the quantity of a good that consumers wish to purchase at that price. We consider:

$$D(p) = \frac{A}{p^\eta}, \quad p > 0,$$

where $A > 0$ is a scale parameter and $\eta > 0$ is the **price elasticity of demand** (a higher η means consumers are more sensitive to price changes).

1. Using the **sequential definition** of continuity, show that D is continuous on $(0, \infty)$: if $p_n \to p$ with $p > 0$, then $D(p_n) \to D(p)$. You may need to use the following theorem:

> Theorem
>
> If $f : K \to V$ is continuous and $g : V \to S$ is continuous, then $g \circ f$ is continuous.

2. Let $\eta = 1$, $A = 1$. Construct an explicit sequence $p_n \to 0^+$ and show $D(p_n) \to +\infty$. What does this say about market behavior when prices approach zero?
3. Is D uniformly continuous on $(0, \infty)$? On $[\underline{p}, \bar{p}]$ for fixed $0 < \underline{p} < \bar{p}$? Justify briefly.

Sequences

Convergence of a Sequence

A sequence $(a_n)_{n\geq 1}$ **converges to** $L \in \mathbb{R}$, written $a_n \to L$, if:

$$\forall\, \varepsilon > 0,\ \exists\, N \in \mathbb{N} :\ n > N \implies |a_n - L| < \varepsilon.$$

Think of it as a game: your opponent picks any tolerance $\varepsilon > 0$, however tiny, and you must produce a threshold N after which every term of the sequence lies within ε of L.

Exercise 3 : Convergence from the definition

Let $a_n = \dfrac{3n+1}{n+2}$.

1. Compute the first five terms of the sequence and conjecture the limit L.
2. Prove your conjecture rigorously using the ε-N definition. *(Hint: simplify* $|a_n - L|$ *algebraically first.)*
3. Find explicitly the smallest $N(\varepsilon)$ such that $n > N(\varepsilon)$ guarantees $|a_n - L| < \varepsilon$.

Exercise 4 : Monotone convergence and the harmonic series (difficult)

Monotone Convergence Theorem

A sequence that is **increasing** ($a_{n+1} \geq a_n$) and **bounded above** converges. A sequence that is **decreasing** ($a_{n+1} \leq a_n$) and **bounded below** converges. You do *not* need to know the limit in advance.

Consider the two sequences of partial sums:

$$S_n = \sum_{k=1}^{n} \frac{1}{k^2}, \qquad H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

1. Show that both (S_n) and (H_n) are strictly increasing.
2. Show that (S_n) is bounded above. *(Hint: show* $\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$ *for* $k \geq 2$*, then telescope.)*
3. Conclude by the Monotone Convergence Theorem that (S_n) converges. (Its limit is $\pi^2/6$, a famous result you may use without proof.)
4. Show that (H_n) is **not** bounded above, and hence diverges. *(Hint: group terms as* $H_{2^n} \geq 1 + n/2$*, by noting that each doubling block contributes at least* $1/2$*.)*
5. What does the divergence of (H_n) tell you about the statement "each individual term $1/k$ goes to zero"? Is this sufficient for a series to converge?

Exercise 5 : A price adjustment process

Fixed Point

A **fixed point** of a function h is a value x^* such that $h(x^*) = x^*$. In economics, fixed points often represent equilibria: states that are self-sustaining once reached.

A market for a commodity adjusts prices over time according to the rule:

$$p_{n+1} = \frac{1}{3}p_n + 8, \qquad p_0 = 0.$$

Think of p_n as the market price in period n: each period, the price partially reverts toward a long-run value of 12.

1. Compute p_1, p_2, p_3, p_4, p_5. Is the sequence increasing? (prove or disprove)
2. Show the sequence is bounded above by 12. *(Hint: show by induction that $p_n \leq 12$ for all n.)*
3. Conclude using the Monotone Convergence Theorem that (p_n) converges to some limit p^*.
4. Find p^* by taking limits on both sides of the recursion. Interpret p^* as the long-run equilibrium price.

Rates of Convergence

Big-O and Little-o Notation

Let (a_n) and (b_n) be sequences with $b_n > 0$.

- $a_n = O(b_n)$: there exists $C > 0$ and N such that $n > N \Rightarrow |a_n| \leq C b_n$. We say a_n is *at most of order* b_n.
- $a_n = o(b_n)$: $a_n / b_n \to 0$. We say a_n is *negligible relative to* b_n.

Key relationship: $o(b_n) \Rightarrow O(b_n)$, but not conversely.

Exercise 6 : Classifying rates

Classify each sequence below as $O(n^{-1})$, $o(n^{-1})$, $O(n^{-1/2})$, $o(n^{-1/2})$, or none of these, as $n \to \infty$. A sequence may belong to several categories. Justify every answer.

$$a_n = \frac{2}{n}, \quad b_n = \frac{\ln n}{n}, \quad c_n = \frac{1}{n^{3/4}}, \quad d_n = \frac{1}{\sqrt{n}\,\ln n}, \quad e_n = \frac{1}{n^2}$$

Exercise 7 : Taylor remainder and approximation quality

Consider the function $f(x) = \sqrt{1+x}$, expanded around $x = 0$.

1. Compute the first-order Taylor expansion $P_1(x)$ of f around $x = 0$.
2. Show that the remainder $R_1(x) = f(x) - P_1(x)$ satisfies $R_1(x) = o(x)$ as $x \to 0$. *(This is exactly what Taylor's theorem promises.)*
3. Show that $R_1(x) = O(x^2)$ as $x \to 0$ but $R_1(x) \neq o(x^2)$. *(Compute $R_1(x)/x^2$ and take the limit.)*
4. A firm's marginal cost is approximated as $MC(q) \approx 1 + \frac{q}{2}$ for small q, where the true marginal cost is $MC(q) = \sqrt{1+q}$. Suppose $q = O(n^{-1/2})$ (the quantity shrinks as the market grows). What is the order of the approximation error $MC(q) - (1 + q/2)$ in terms of n? Do you think this is good enough for an actual firm?

Linearization (Univariate)

Exercise 8 : Bond pricing, duration, and convexity

In financial economics, a **zero-coupon bond** is a contract that pays exactly \$1 at a future date T (called the **maturity**). If the continuously compounded interest rate is $r > 0$, the price of this bond today is:

$$P(r) = e^{-rT}.$$

A lower interest rate means money in the future is more valuable today, so P decreases in r.

1. Compute $P'(r)$ and $P''(r)$. Verify that P is decreasing and convex in r.

2. Write the second-order Taylor expansion of $P(r)$ around a given rate r_0:

   $$P(r) \approx P(r_0) + P'(r_0)(r - r_0) + \frac{1}{2}P''(r_0)(r - r_0)^2.$$

   Show that this can be written as:

   $$P(r) \approx P(r_0)\left[1 - T(r - r_0) + \frac{T^2}{2}(r - r_0)^2\right].$$

   The coefficient T is called the **duration** of the bond.

3. Interpret the linear term: if r rises by 1 percentage point (i.e. $r - r_0 = 0.01$), by approximately how much does the bond price fall, in percentage terms? Why does a longer-maturity bond fall more?

4. The second-order term captures **convexity**: the price curve bends upward. Show that the true price $P(r)$ is always **above** the linear approximation for r around r_0. You will need to show that that $e^u > 1 + u$ for all $u \neq 0$. *(Hint: MVT could prove useful)*

5. What does convexity imply for an investor who holds bonds and faces uncertain interest rates? Is convexity good or bad for the investor? Explain in one or two sentences.

Multivariate Calculus

Gradient and Jacobian

Let $f : \mathbb{R}^n \to \mathbb{R}$ be C^1. The **gradient** of f at x^* is the vector of partial derivatives:

$$\nabla f(x^*) = \left(\frac{\partial f}{\partial x_1}(x^*), \dots, \frac{\partial f}{\partial x_n}(x^*) \right)^\top .$$

For $F = (f_1, \dots, f_m) : \mathbb{R}^n \to \mathbb{R}^m$, the **Jacobian** is the $m \times n$ matrix:

$$J_F(x^*) = \begin{pmatrix} \partial f_1/\partial x_1 & \cdots & \partial f_1/\partial x_n \\ \vdots & \ddots & \vdots \\ \partial f_m/\partial x_1 & \cdots & \partial f_m/\partial x_n \end{pmatrix}_{x=x^*} .$$

Exercise 9 : Gradient and isoprofit curves (difficult)

A firm's profit is:

$$\pi(K, L) = K^{1/3} L^{1/3} - wL - rK,$$

where $K > 0$ is capital, $L > 0$ is labor, $w > 0$ is the **wage** (cost per unit of labor), and $r > 0$ is the **rental rate of capital** (cost per unit of capital). These are fixed parameters, not variables.

1. Compute the gradient $\nabla \pi(K, L)$ at a general point (K, L).
2. Evaluate $\nabla \pi$ at the specific point $(K, L) = (1, 1)$ when $w = r = \frac{1}{3}$.
3. An **isoprofit curve** is the set of all (K, L) pairs that give the same profit level $\bar{\pi}$:

$$\mathcal{I} = \{(K, L) \in \mathbb{R}^2_{++} : \pi(K, L) = \bar{\pi}\}.$$

   Without doing any computation, explain why $\nabla \pi$ is said to be **orthogonal** to $\mathcal{I}$ at any point on it. *(Hint: think about what it means to move along $\mathcal{I}$.)*

4. At the point $(1, 1)$ with $w = r = 1/3$, you found $\nabla \pi(1, 1) = (0, 0)^\top$. This means $(1, 1)$ is a **critical point** of π: the gradient vanishes, and the standard orthogonality argument breaks down (the zero vector is orthogonal to everything, which gives no useful information about the geometry). To see the gradient-isoprofit orthogonality, move to a nearby point where the gradient is nonzero. Take $(K, L) = (2, 1)$ with the same $w = r = 1/3$.
   (a) Compute $\nabla \pi(2, 1)$.
   (b) The isoprofit curve through $(2, 1)$ is implicitly defined by $\pi(K, L) = \pi(2, 1)$. Differentiate this constraint implicitly with respect to K to find $\frac{dL}{dK}$ at $(2, 1)$, i.e. the slope of the tangent to the isoprofit curve.
   (c) Write down the tangent direction vector $v = (1, dL/dK)$ and verify by direct computation that $\nabla \pi(2, 1) \cdot v = 0$.

Exercise 10 : Jacobian of an excess demand system

When consumers want to buy more of a good than is available, we say there is **excess demand**; when there is more available than consumers want, there is **excess supply**. A market is in **equilibrium** when neither occurs: the quantity consumers wish to buy exactly equals the quantity supplied. We call this an equilibrium price because, once reached, there is no pressure for the price to change: every seller finds a buyer at that price, and every buyer finds a seller.

Consider the following two-good excess demand system:

$$Z_1(p_1, p_2) = \frac{p_2}{p_1} - 1, \qquad Z_2(p_1, p_2) = \frac{p_1}{p_2} - 1.$$

Here $Z_i > 0$ means consumers want more of good i than is available (excess demand); $Z_i < 0$ means there is a surplus.

1. Verify that $(p_1, p_2) = (1, 1)$ is an equilibrium, i.e. $Z_1(1, 1) = Z_2(1, 1) = 0$.
2. Compute the Jacobian $J_Z(p_1, p_2)$ and evaluate it at $(1, 1)$.
3. Interpret each entry of $J_Z(1, 1)$ economically. In particular: what does $\partial Z_1/\partial p_2 > 0$ tell you about how the two goods are related?
4. Compute the directional derivative of Z_1 at $(1, 1)$ in the direction $v = (1, 1)/\sqrt{2}$ (a simultaneous equal rise in both prices). Interpret the result: what happens to excess demand for good 1 when all prices scale up equally?
5. Verify that $p_1 Z_1(p_1, p_2) + p_2 Z_2(p_1, p_2) = 0$ for all (p_1, p_2). This is known as **Walras' Law**: the total value of excess demand across all goods is always zero. Give an economic interpretation: why does it make sense that consumers, taken together, cannot be simultaneously trying to buy more than they can afford?

Exercise 11 : Chain rule and the Solow growth decomposition (difficult)

In macroeconomics, a **production function** maps inputs to output. The **Cobb-Douglas** production function is widely used:

$$Y(K, L) = K^\alpha L^{1-\alpha}, \quad \alpha \in (0, 1),$$

where Y is output (GDP), K is capital, L is labor, and α is capital's share of income.

Both K and L grow over time:

$$K(t) = K_0\, e^{g_K t}, \qquad L(t) = L_0\, e^{g_L t},$$

where $g_K > 0$ and $g_L > 0$ are the growth rates of capital and labor respectively.

1. Use the **chain rule** to compute $\dfrac{dY}{dt}$ as a function of $K(t)$, $L(t)$, g_K, g_L, and α.
2. Define the **growth rate of output** as $g_Y = \dfrac{\dot{Y}}{Y} = \dfrac{1}{Y}\dfrac{dY}{dt}$. Show that:
$$g_Y = \alpha\, g_K + (1 - \alpha)\, g_L.$$
3. This result is the **Solow growth decomposition** (after Nobel laureate Robert Solow, who developed it in the 1950s). Interpret it as follows: in a competitive economy, firms pay each input its marginal product, so the wage bill is $(1 - \alpha)Y$ and the capital income is αY. Explain why α and $(1 - \alpha)$ appear as weights, and what the decomposition says about how growth in each input contributes to output growth.
4. Suppose capital grows at $g_K = 3\%$ and labor at $g_L = 1\%$, with $\alpha = 1/3$. Compute the implied output growth rate g_Y. Break down the contribution of each factor explicitly.
5. In practice, observed output growth is often **higher** than $\alpha g_K + (1 - \alpha) g_L$. Solow called this gap **Total Factor Productivity (TFP)** growth, or the **Solow residual**. If actual output growth is $g_Y = 2.5\%$ using the numbers from Question 4, what is the TFP growth rate? What does TFP capture that capital and labor growth do not?

Exercise 12 : Implicit function theorem: money market equilibrium

Implicit Function Theorem (scalar case)

Let $F : \mathbb{R}^2 \to \mathbb{R}$ be C^1, and suppose $F(x_0, y_0) = 0$ with $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$. Then near (x_0, y_0) there exists a unique C^1 function $y = g(x)$ such that $F(x, g(x)) = 0$, and:

$$g'(x_0) = -\left.\frac{\partial F/\partial x}{\partial F/\partial y}\right|_{(x_0,y_0)}.$$

The **money market** is in equilibrium when money demand equals money supply. A standard specification is:

$$M = P \cdot L(Y, i),$$

where:

- $M > 0$ is the (fixed) **money supply**, controlled by the central bank
- $P > 0$ is the **price level**
- $Y > 0$ is **real output** (GDP)
- $i > 0$ is the **nominal interest rate**
- $L(Y, i) = \frac{Y}{i}$ is **real money demand**: higher output raises demand, higher interest rates reduce it (since bonds become more attractive)

Rewrite the equilibrium condition as:

$$F(Y, i;\, M, P) \equiv \frac{PY}{i} - M = 0.$$

A note on realism. The functional form $L(Y, i) = Y/i$ is a deliberate simplification. Empirical money demand functions typically take the form $L(Y, i) = Y^\beta / i^\gamma$ with $\beta, \gamma > 0$, and the log-linearized version $\ln L = \beta \ln Y - \gamma \ln i$ is standard in monetary economics. The unit elasticities ($\beta = \gamma = 1$) assumed here make the algebra clean while preserving the correct qualitative properties: money demand rises with output and falls with the interest rate. The IFT results below are qualitatively robust to more general specifications. Moreover, for the sake of simplicity, we will work with $i = 1$ (a 100% interest rate)...

1. Verify that the point $(Y_0, i_0) = (M/P, 1)$ satisfies $F = 0$.
2. Check that the IFT condition $\frac{\partial F}{\partial i} \neq 0$ holds at (Y_0, i_0). Why is this the relevant partial derivative? *(Think about which variable we want to express as a function of which.)*
3. Apply the IFT to compute $\dfrac{\partial i^*}{\partial Y}$: how does the equilibrium interest rate respond to a rise in output? This relationship traces out the **LM curve**.
4. Apply the IFT to compute $\dfrac{\partial i^*}{\partial M}$: how does the interest rate respond to an increase in the money supply? Provide economic intuition.
5. *(Harder.)* Now treat both Y and M as varying simultaneously. Use the IFT formula

$$\begin{pmatrix} \partial i^*/\partial Y \\ \partial i^*/\partial M \end{pmatrix} = -\left(\frac{\partial F}{\partial i}\right)^{-1} \begin{pmatrix} \partial F/\partial Y \\ \partial F/\partial M \end{pmatrix}$$

to recover your answers from parts 3 and 4 in a single computation. *(Note: since F maps to $\mathbb{R}$, the "matrix" $\partial F/\partial i$ is just a scalar, and its inverse is simply its reciprocal. This is the scalar special case of a more general matrix IFT you will encounter later in the course.)*